

BUS 713 Financial Analytics

Assessment 1

Financial Data Analysis & Insight Report

You are working as a financial analyst at an investment firm. Interest in large-cap technology stocks has increased significantly with advances in artificial intelligence, and clients are asking whether investing in individual firms offers better opportunities than tracking the broader market.

You have been asked to evaluate the performance of selected large-cap stocks relative to the Nasdaq index for the period between January 1, 2005 and December 31, 2025. Students may select any two stocks of their choice, excluding Apple and Microsoft. The focus is not only on returns, but on how those returns relate to risk and inflation over time.

The analysis should present returns for the selected stocks and the Nasdaq index. Returns should be computed at an appropriate frequency and visualised over time. Graphs of prices and returns should be included and briefly discussed to highlight key patterns.

Students should analyse risk, focusing on the variability of returns. Volatility should be computed and compared across assets, and the behaviour of volatility over time should be considered. Periods of calm and turbulence, as well as evidence of volatility clustering, should be identified and interpreted.

The analysis should then consider downside risk. Measures such as Value at Risk (VaR) and Expected Shortfall (ES) should be computed using the return series. These measures should be interpreted clearly, with attention to what they reveal about potential losses and extreme outcomes.

Finally, the analysis should evaluate long-run performance. Growth over the sample period should be summarised using compound annual growth rate (CAGR) for each stock and the Nasdaq. Returns should also be interpreted in both nominal and real terms, with inflation-adjusted CAGR calculated using US CPI data. The impact of inflation on long-term investment performance should be clearly discussed.

The final output should provide a clear evaluation of how the selected stocks compare to the Nasdaq, taking into account return, risk, and inflation, and reflecting on what these results imply for investment decisions.

Submission Components

1. Video Presentation

You must submit a 6–10-minute recorded video presentation aimed at management and other stakeholders.

Part A: Code and Approach

- Walk through your R code
- Explain what you are doing and why

Part B: Insights and Interpretation

- Present your key findings
- Explain what do they mean for an investor.

You must:

- be visible on camera throughout
- show your student ID at the start

2. R script

It should be well structured and reproducible. Comments should explain the key steps.

3. Executive Summary

An executive summary of your work, the challenges you faced in completing this exercise, and how you might apply the tools and techniques you used in this exercise in your work and life.

The summary should briefly answer the following questions:

- What did you analyse?
- What did you find?
- What would you recommend?
- What are the limitations?

Your executive summary should be approximately 250 words.